

Telegram Test Failure Analysis & Production Fixes

Project: Vyapar AI Employee | Tenant: HONS | Date: June 3, 2026 | Source: Live Telegram screenshots

A. Execution Paths (Failed Messages)

"Nepal ma internet ko user kati chha?"

```
user message
→ detect_intent()          → "pricing" (BUG: bare "kati" keyword)
→ route_turn()              → general_knowledge (FIXED)
→ _resolve_sales_mode()     → True via lead_stage=interested (BUG)
→ lead_block injected      → 300 Mbps from active lead
→ LLM reply                 → package pitch appended
→ validate_response()       → overlap → build_fallback_reply("sales")
→ output                    → "package suggest garna sakchha"
```

"Tapai le kati padhnu bhayeko chha?"

```
user message → detect_intent() → "pricing" (BUG)
→ route_turn() → general_knowledge (FIXED)
→ _resolve_sales_mode() → True via warm lead (BUG)
→ fallback → sales package text
```

"Mero naam Khemraj Adhikari ho"

```
user message → extract_memory_facts() → name saved
→ upsert_lead() still runs → old requested_speed=300 Mbps merged
→ lead_block in prompt → "300 Mbps interest" (BUG)
→ LLM → "Tapainko 300 Mbps ko chaso chha"
```

"Nepali type garnus na" / emoji 🤔

```
user message → detect_language() → "english" on emoji (BUG)
→ resolve_session_language() → overwrites nepali → english
→ LLM → English reply / "I only speak English" contradiction
```

Coverage check repeated

```
user message → turn=sales → sales_mode=True + COVERAGE_SALES_HINT every turn
→ coverage_mention_count not tracked (BUG)
→ validate_response fallback → generic/sales text again
```

"secondary router ko kati parchha?"

```
user message → detect_intent() → "pricing"
→ route_turn() → sales → product_block → 100 Mbps catalog
→ LLM → package pitch instead of "price unknown"
```

B. Module Blame Matrix

Failure	Primary Module	Secondary
General questions → sales	intent_engine.py	ai_employee_engine.py _resolve_sales_mode
Warm lead forces sales on any turn	ai_employee_engine.py	turn_router.py
Name turn → 300 Mbps	ai_employee_engine.py	lead_extractor.py
Language flip	language_lock.py	ai_employee_engine.py
Coverage loop	session_state_manager.py	response_validator.py
Unknown router price → pitch	turn_router.py	response_validator.py

C. Bugs & Patches Applied

BUG-1 Intent routing

Root cause: `detect_intent()` treats any "kati" as pricing.

File: `intent_engine.py` — `detect_intent`, `_is_general_knowledge`, `_is_pricing_intent`

Fix: General-knowledge patterns checked first; "kati" only pricing with `parchha/ho/cha` or product context.

BUG-2 Sales mode on non-sales turns

Root cause: `_resolve_sales_mode()` did not block all non-sales turn types when `lead_stage=interested`.

File: `ai_employee_engine.py` — `_resolve_sales_mode`

Fix: `NON_SALES_TURNS` frozenset returns False immediately for greeting, general_knowledge, memory_write, correction, etc.

BUG-3 Name → 300 Mbps contamination

Root cause: Name-only turn still ran `upsert_lead()` and injected `lead_block` with prior `requested_speed`.

Files: `turn_router.py` (`memory_write` turn), `ai_employee_engine.py` (`suppress_lead_context`), `memory_validator.py` (skip package on name-only)

BUG-4 Language lock failure

Root cause: Language replaced every turn; emoji flipped Nepali → English; LLM claimed "English only" in Nepali.

File: `language_lock.py` — sticky lock, preference detection, stronger prompt

BUG-5 Coverage repetition

Root cause: No coverage counter; `COVERAGE_SALES_HINT` on every sales turn.

Files: `session_state_manager.py`, `memory_db.py`, `response_validator.py`, `turn_router.py`

BUG-6 Unknown product → package pitch

Root cause: No `unknown_product` route.

File: `turn_router.py` — `UNKNOWN_PRODUCT_PATTERNS` → direct "confirmed price chaina"

BUG-7 Bad validator fallback

Root cause: `build_fallback_reply()` returned sales text on general_knowledge validation failure.

File: `response_validator.py` — turn-specific non-sales fallbacks

D. Logging Added

Event	Location
INTENT_DETECTED	ai_employee_engine.py
TURN_ROUTED	ai_employee_engine.py
MEMORY_READ	session_state_manager.log_memory_read
MEMORY_WRITE	session_state_manager.log_memory_write
LANGUAGE_LOCK_APPLIED	language_lock.py, ai_employee_engine.py
RESPONSE_VALIDATED	response_validator.py

Changed Files

```
intent_engine.py
turn_router.py
language_lock.py
memory_validator.py
response_validator.py
session_state_manager.py
memory_db.py
ai_employee_engine.py
tests/test_p0_phase3.py
docs/migrations/002_language_lock_coverage.sql
```

Database Migration

```
ALTER TABLE conversation_state ADD COLUMN IF NOT EXISTS language_locked BOOLEAN DEFAULT FALSE;
ALTER TABLE conversation_state ADD COLUMN IF NOT EXISTS coverage_mention_count INTEGER DEFAULT 0;
```

Expected After Fix

User Message	Expected Reply
Nepal ma internet ko user kati chha?	"confirmed data chaina" — no Mbps
Tapai le kati padhnu bhayeko chha?	AI employee answer — no package
Mero naam Khemraj Adhikari ho	"Naam save bhayo" — no 300 Mbps
Nepali type garnus na	Nepali lock; stays Nepali
secondary router ko kati parchha?	"confirmed price chaina" — no pitch
Coverage asked 3rd time	No repeat coverage block

Deploy Checklist

1. Deploy all changed files atomically to Render
2. Run migration on Postgres if table already exists
3. Restart service
4. Run: `python -m unittest discover -s tests -v`
5. Re-test the 6 messages above in Telegram

